

Jiayi Zhang

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Education

University of Nottingham Ningbo China (UNNC)

BSC (HONS) MATHEMATICS WITH APPLIED MATHEMATICS

- GPA - 3.83/4.00

Ningbo China

Sep 2024 - Jun 2028 (Expected)

Research Interests

My research focuses on aligning **learning** and **control** to build intelligent systems that **perceive**, **reason**, and **interact** with the physical world. Grounded in **mathematics**, **AI**, and **control theory**, I study how large-scale **multimodal models** integrate with **decision-making frameworks** to go beyond passive perception, toward **embodied intelligence** operating in complex environments.

Experience

Tsinghua University (THU)

RESEARCH ASSIST AT [TEA LAB](#), ADVISED BY [PROF. HUAZHE XU](#)

- Research Topics:** Robotics, Embodied AI

Shanghai Jiao Tong University (SJTU)

RESEARCH ASSIST AT [IWIN-FINS LAB](#), ADVISED BY [PROF. JIANPING HE](#)

- Research Topics:** generative model, grasping, equivariant and invariant

The University of Hong Kong (HKU)

RESEARCH INTERN AT [IoT LAB](#), ADVISED BY [PROF. EDITH C.H. NGAI](#)

- Research Topics:** federated learning, fairness, spatio-temporal graphs.

Beijing, China

October 2025

Shanghai, China

June 2025 - Present

Hong Kong SAR, China

March 2025 – August 2025

Publications

PREPRINTS

[P2] F²STNet: Fair and Federated Spectral-Temporal Modeling for Graph Forecasting

Jiayi Zhang*, Jinfeng Xu*, Hewei Wang*, Siyuan Cen, Haidong Huang, Yiyao Zhan, Zheyu Chen, Jinjiang You, Ai Jian, Edith C. H. Ngai

Under review, 2025

[P1] Multi-modal Dynamic Proxy Learning for Personalized Multiple Clustering

Jinfeng Xu, Zheyu Chen, Shuo Yang, Jinze Li, Ziyue Peng, Zewei Liu, Hewei Wang, [Jiayi Zhang](#), Edith C. H. Ngai

Under review, 2025

CONFERENCE PROCEEDINGS

[C3] Multi-Cali Anything: Dense Feature Multi-Frame Structure-from-Motion for Large-Scale Camera Array Calibration

Jinjiang You*, Hewei Wang*, Yijie Li, Mingxiao Huo, Long Van Tran Ha, Mingyuan Ma, Jinfeng Xu, [Jiayi Zhang](#), Puzhen Wu, Shubham Garg,

Wei Pu

IROS, 2025 (Oral) [\[Paper\]](#)

[C2] Spec-LLAVA: Accelerating Vision-Language Models with Dynamic Tree-Based Speculative Decoding

Mingxiao Huo*, [Jiayi Zhang](#)*, Hewei Wang*, Jinfeng Xu, Zheyu Chen, Huilin Tai, Ian Yijun Chen

ICML Workshop, 2025 [\[Paper\]](#)

[C1] Segregation and Context Aggregation Network for Real-Time Cloud Segmentation

Yijie Li*, Hewei Wang*, [Jiayi Zhang](#), Jinjiang You, Jinfeng Xu, Puzhen Wu, Yunzhong Xiao, Soumyabrata Dev

ICLR Workshop, 2025 [\[Paper\]](#)

JOURNAL ARTICLES

[J3] FGSCare: A Feature-driven Grid Search-based Machine Learning Framework for Coronary Heart Disease Prediction

Zhenqi Li*, [Jiayi Zhang](#)*, Yuchen Wang*, Jinjiang You, Junhao Zhong, Soumyabrata Dev

Qeios preprint, 2025 [\[Paper\]](#)

[J2] TrafficKAN-GCN: Graph Convolutional-based Kolmogorov-Arnold Network for Traffic Flow Optimization

[Jiayi Zhang](#)*, Yiming Zhang*, Yuan Zheng, Yuchen Wang, Jinjiang You, Yuchen Xu, Wenxing Jiang, Soumyabrata Dev

arXiv preprint, 2025 [\[Paper\]](#)

[J1] **RAINER: A Robust Ensemble Learning Grid Search-Tuned Framework for Rainfall Patterns Prediction**
Zhenqi Li*, Junhao Zhong*, Hewei Wang, Jinfeng Xu, Yijie Li, Jinjiang You, Jiayi Zhang , Runzhi Wu, Soumyabrata Dev
arXiv preprint, 2025 [[Paper](#)]

Honors & Awards

2022 **Silver Medal**, Chinese Mathematical Olympiad (CMO), 2022

Shenzhen, China

Leadership/Mentorship/Social

Co-Principal Investigator

UNIVERSITY OF NOTTINGHAM NINGBO CHINA (UNNC)

Ningbo China

Summer 2025-Fall 2027

- Developed a 30B multimodal model for academic manuscript review and feedback support in university settings.
- In-progress internal project for student-faculty support.

Skills

Programming Languages Python, C/C++, $\LaTeX$, Java, MATLAB, HTML, CSS, JavaScript, Kotlin

Frameworks PyTorch, Tensorflow, NumPy, Flask, MySQL, Git, Anaconda, OpenCV, ROS

Machine Learning Models CNN, ResNet, Faster R-CNN, YOLO, LSTM, Transformer, BERT, ViT, GANs, VAE, Stable Diffusion

Platforms Unitree Z1 Pro, Franka Research 3